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RESEARCH ARTICLE

Self-Supervised DRL With Twin Critics: A Novel Framework for Glioblastoma Survival Prognostication

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ABSTRACT

The accurate survival prediction in glioblastoma patients remains a major challenge, largely owing to the considerable variability among brain tumor subtypes and their clinical profiles. This research introduces an innovative SimCLR-Twin Critic DDPG framework that integrates self-supervised representation learning with deep reinforcement learning for enhanced prognostic prediction. Tumor subregion segmentations were independently obtained using four advanced models namely nnU-Net, VNet, SwinUNETR, and UNet. Based on the performance of these segmentation models, deep features were extracted from the nnU-Net segmented tumor subregions and combined with handcrafted radiomics features. The combined features were subjected to the feature selection process using BorutaShap, followed by self-supervised pretraining through SimCLR to improve feature representation and generalization. The optimized features were then fed into a Twin Critic DDPG agent designed for regression-based survival time predictions. The proposed method outperformed existing techniques on the BraTS 2020 dataset, achieving the lowest MSE of 31 062.36 and the highest C-index of 0.87 in overall survival prediction. To further confirm robustness and generalizability, the framework was externally validated on the BraTS 2019 dataset, where it achieved a comparable MSE of 31 156.56 and a C-index of 0.86 using fused features. Additionally, LIME-based local interpretability provided clinically relevant explanations for individual predictions, thereby enhancing trust in the AI-driven system. This study highlights the SimCLR-Twin Critic DDPG framework as a robust and interpretable solution for accurate survival prediction in glioblastoma patients.

1 | Introduction

Glioblastoma (GBM), categorized as a Grade IV glioma by the World Health Organization (WHO), is recognized as the most aggressive and fatal primary brain tumor in adults [1]. It is characterized by rapid progression, high recurrence rates, and an extremely poor prognosis. Despite advances in surgical resection, radiotherapy, and chemotherapy, current treatment strategies typically extend patient survival to less than 2 years [2]. Although traditional prognostic factors such as age, extent of resection and Karnofsky Performance Status (KPS) remain important, they often do not fully capture the complex biological and spatial heterogeneity of GBM [3].

In recent years, the emergence of multimodal Magnetic Resonance Imaging (MRI) has enabled a more comprehensive, non-invasive assessment of tumor structure, progression, and treatment response [4]. Various MRI modalities (e.g., T1, T1CE, T2, and FLAIR) provide complementary information about edema, necrosis, and contrast-enhancing regions [5]. In this

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regard, radiomics, the process of extracting quantitative features from medical images, has proven to be an effective approach for capturing both the tumor phenotype and its biological heterogeneity [6, 7]. When integrated with machine learning (ML) and deep learning (DL), it enhances its capability to produce data-driven prognostic models that learn complex patterns linked to survival outcomes [8–11].

Recent studies have increasingly explored the fusion of radiomics with deep learning derived features to enhance prognostic accuracy in glioblastoma [12, 13]. However, these combined features may introduce high dimensionality and heterogeneity, which can compromise model generalizability and interpretability, particularly when dealing with the limited data. Additionally, most existing methods rely on static learning paradigms, limiting their adaptability to evolving clinical scenarios. In contrast, reinforcement learning (RL) offers a promising alternative [14, 15]. Through trial-and-error interactions with the data environment and optimizing long-term rewards, RL models can learn intricate, nonlinear relationships, even when working with smaller datasets. This work presents a novel reinforcement learning-based survival prediction framework that integrates contrastive representation learning with deep RL for glioblastoma patients. Specifically, SimCLR, a self-supervised learning technique is employed to extract robust and discriminative feature representations from fused radiomics and deep features. These compact representations help mitigate issues related to high dimensionality and limited data. The survival prediction task is then formulated as a continuous valued regression problem using a Twin Critic Deep Deterministic Policy Gradient (DDPG) agent, which enables the model to capture the complex, nonlinear dynamics of survival outcomes. This work marks the first attempt to implement a deep RL framework for the direct prediction of continuous overall survival in GBM patients, utilizing fused imaging features learned through self-supervised contrastive learning. The core contributions of this research can be summarized as.

- This paper proposes a SimCLR-Twin Critic DDPG framework for continuous survival prediction of glioblastoma patients by reformulating survival analysis as a reinforcement learning task capable of capturing complex survival dynamics that are beyond the reach of classical and even modern ML/DL methods.
- A self-supervised SimCLR strategy is used to create robust feature representations while reducing the need for large labeled datasets.
- A multi-level feature fusion method uses deep features from nnU-Net segmentations and handcrafted radiomics features across different MRI modalities to help with tumor characterization and prognostic predictions.
- The segmentation outputs from nnU-Net, VNet, SwinUNETR, and U-Net are all collected to ensure reliable segmentation, from which the best-performing model is applied for deep feature extraction.
- The framework is made more interpretable and clinically reliable with explainability tools such as BorutaShap for global feature selection and LIME for local interpretability.
- The tests run show that the proposed framework outperforms state-of-the-art ML/DL baselines, which achieve lower MSE and higher prognosis accuracy.

The rest of this paper unfolds in the following manner. Section 2 examines prior work relevant to survival prediction. Section 3 outlines the proposed methodology, covering dataset details, multimodel segmentation, feature extraction, SimCLR-based self-supervised learning and the Twin Critic DDPG framework. Section 4 then presents the experimental setup, the evaluation results, and a comparison against existing methods. In conclusion, Section 5 summarizes the key findings of this work.

2 | Related Works

Recent developments in the field of AI-enhanced glioblastoma survival prediction have increasingly emphasized the integration of radiomics, deep learning, and self-supervised learning to improve prognostic precision. Chen et al. proposed a subregion-specific AI framework for glioblastoma survival analysis [16]. Their model extracted both 8231 radiomic and geometric features from the 117 GBM patients, followed by the application of a feature space optimization algorithm to achieve optimal alignment between MRI sequences and tumor subregions. It selects 235 features with K-means clustering and employs Support Vector Classification (SVC) and sub-regression ensemble models to predict 1 year and overall survival. The study emphasized the value of combining optimized feature selection with advanced imaging to improve prognostic models. Expanding on segmentation-driven prediction frameworks, Tran et al. introduced an approach centered on FLAIR MRI images, integrating tumor segmentation, local-global feature learning, and self-supervised learning for survival prediction [17]. Their method employed a LightGBM ensemble with majority voting, leveraging patch-wise relationships within imaging data to reduce overfitting and demonstrating the value of self-supervised techniques for improving survival prediction accuracy in glioblastoma.

Building upon this radiomics-driven survival prediction methodology, Rajput et al. proposed a detailed AI model that combines 3D U-Net tumor segmentation with the extraction of 1265 radiomics and clinical features [18] To improve model interpretability and efficiency, they employed recursive feature elimination and permutation importance to select 29 significant predictors. A random forest regressor was used for survival prediction, further enhanced by interpretability tools. In a parallel effort to advance multimodal MRI applications, Zhu et al. introduced an automated framework that utilizes a modified 3D U-Net model for tumor subregion segmentation, followed by radiomics feature extraction and dimensionality reduction via Principal Component Analysis (PCA) [19]. The optimized features, combined with clinical data, were processed using a Support Vector Regression (SVR) model for overall survival prediction, offering a robust and non-invasive tool for personalized prognosis.

Further reinforcing the value of radiomics pipelines, Baid et al. implemented a similar workflow by extracting 678 features

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from multimodal MRI images after 3D U-Net-based tumor segmentation [20]. These features were then analyzed using XGBoost and Multi-Layer Perceptron (MLP) models, achieving strong predictive accuracy and emphasizing the effectiveness of integrating multimodal features for reliable glioblastoma survival prediction. Moving beyond conventional segmentation networks, Huang et al. introduced an enhanced NLSE-VNet model with integrated attention modules for improved tumor segmentation [21]. Radiomics features extracted from segmented regions were fused with deep features from a CNN and dimensionality reduction was performed via factor analysis. These multimodal features were combined with clinical variables and input into a Random Forest model, demonstrating enhanced OS prediction performance through effective multimodal feature fusion.

Akbar et al. addressed the need for lightweight architectures in the clinical field by proposing a survival prediction model that integrates MobileNet (V1 and V2) with a linear survival prediction model (SPM) [22]. By utilizing a modified MobileNet V2 with frozen layers, they achieved superior accuracy and validated the potential of efficient deep learning frameworks for clinical survival prediction. A deep learning model developed by Yoon et al. leverages convolutional neural networks (CNN) to combine radiomics MRI features and clinical data for survival prediction in 118 glioblastoma patients [23]. Finally, on this progression of AI-driven survival prediction frameworks, Wankhede et al. introduced a comprehensive deep learning-based model combining 3D U-Net and Modified Fuzzy C-Means Clustering (MFCM) for tumor segmentation, Rough Set Theory Grey Wolf Optimizer (RS-GWO) for feature selection and Faster RCNN for survival prediction [24]. This hybrid model achieved high accuracy, surpassing conventional methods in both segmentation and survival forecasting, and underscoring the increasing value of sophisticated feature optimization strategies and ensemble prediction models in precision neuro-oncology.

However, despite these advancements, predicting survival in GBM patients remains a complex challenge. Deep learning often requires large labeled datasets and can overfit on small samples, while conventional ML models struggle to capture the complex, temporal patterns needed for survival prediction. Many approaches also lack transparency for clinical use. Reinforcement Learning especially actor-critic methods offers adaptability to dynamic clinical outcomes. Coupling RL with SimCLR-based self-supervised learning enables robust feature representations from limited data, addressing challenges of data efficiency, adaptability, and interpretability in survival prediction.

3 | Methodology

A new deep reinforcement learning framework is proposed in this study, combining SimCLR-based feature representation learning with a Twin Critic Deep Deterministic Policy Gradient agent to predict continuous overall survival in brain tumor patients. The overall architecture of the proposed pipeline is illustrated in Figure 1.

3.1 | Dataset Description

The benchmark BraTS 2020 and BraTS 2019 datasets are used in this work. They contain multimodal brain MRI scans, which include T1-weighted (T1), T1-weighted contrast enhanced (T1CE), T2-weighted (T2), and Fluid Attenuated Inversion Recovery (FLAIR) sequences [25]. They also consist of expert-annotated segmentation masks for the tumor subregions. The BraTS 2020 dataset consists of 369 glioma cases, of which 237 contain clinical information suitable for survival analysis. This information includes overall survival in days, patient age, and resection status. The BraTS 2019 database similarly consists of 335 glioma cases, 210 of which have survival-related clinical data. For the experiments, only the survival-labeled subsets from both datasets are used, using an 80%–20% split for training and testing, respectively.

3.2 | Image Segmentation

The reliability of survival predictions is fundamentally dependent on the precise segmentation of brain tumor subregions. The exactness of these segmentations is critical, as it directly impacts the quality and discriminative strength of the features extracted for later prediction tasks. To achieve accurate tumor structure delineation, four well-known and high-performing medical image segmentation models such as U-Net, V-Net, SwinUNETR, and nnU-Net were selected [26–29]. These models were chosen for their successful track record in significant medical imaging challenges and their ability to effectively capture both fine details and overall contextual information from volumetric MRI data.

All models used the preprocessed BraTS 2020 multimodal MRI dataset as input, which was utilized to segment clinically significant tumor subregions, such as the Enhancing Tumor (ET), Tumor Core (TC), and Whole Tumor (WT). The performance of the model was measured using the Dice Similarity Coefficient (DSC) and Intersection over Union (IoU). Of the four models, the nnU-Net produced the highest DSC and IoU scores in all tumor subregions. It has strong adaptability and automated optimization. So, nnU-Net was selected as the backbone network for segmentation to extract features for survival prediction.

3.3 | Feature Extraction and Selection

As the nnU-Net model achieved excellent segmentation accuracy, 36 deep features were derived from its delineated tumor subregions. In parallel, radiomics features were independently computed from the segmentation outputs using the PyRadiomics toolkit. A comprehensive set of 105 radiomics descriptors, which included first-order intensity statistics, shape metrics, textural patterns and wavelet-transformed features, was extracted for each tumor subregion across all MRI modalities. The deep features obtained from nnU-Net segmentations were then merged with the handcrafted radiomics features to create a comprehensive, multidimensional feature vector.

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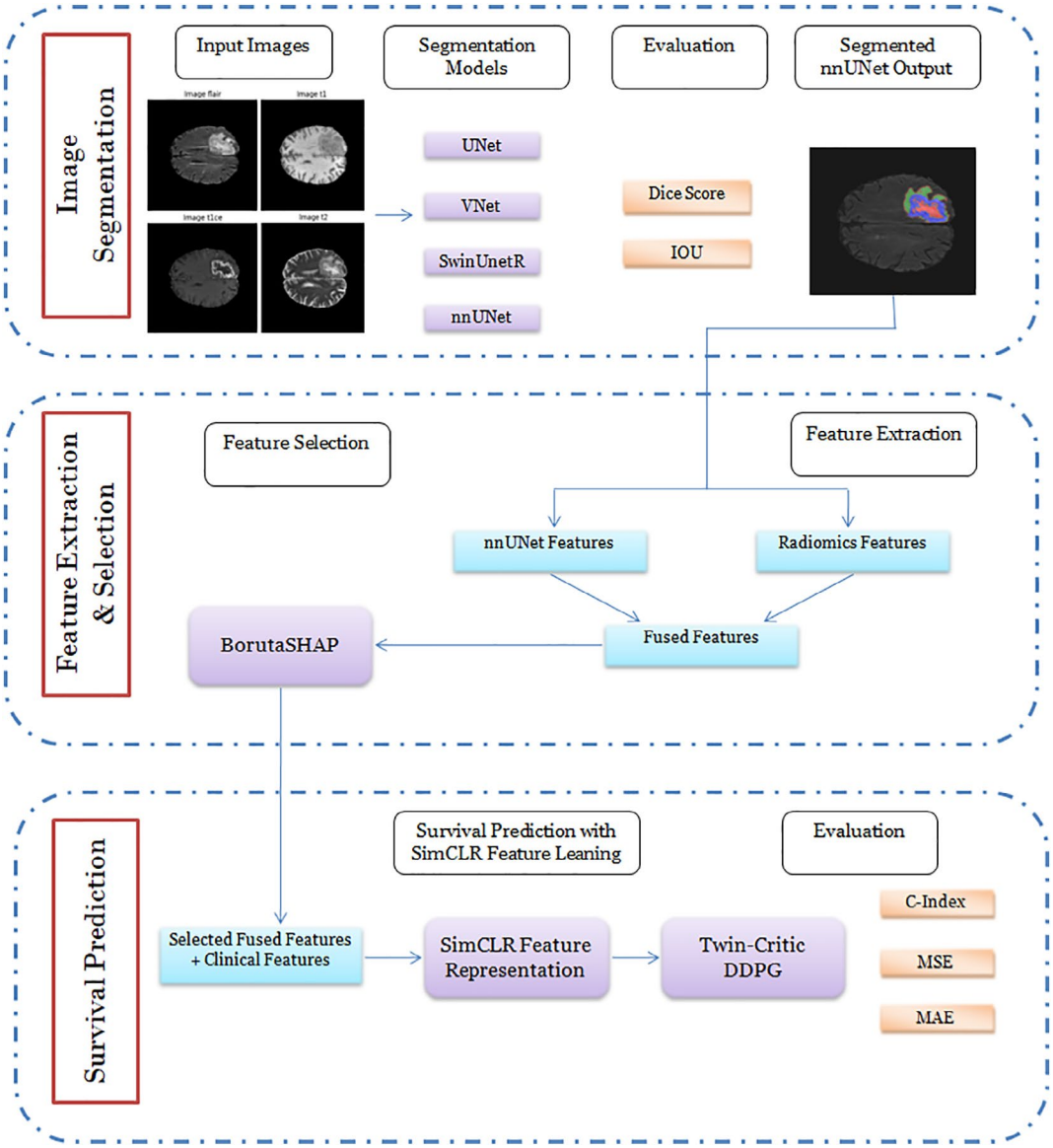

FIGURE 1 | Overview of the proposed SimCLR-Twin Critic DDPG survival prediction framework.

To facilitate dimensionality reduction and improve interpretability, the BorutaShap algorithm, a comprehensive feature selection approach that combines Boruta and SHAP values was employed to identify the 25 most pivotal features that contribute to survival prediction [30]. This approach ensured that both features derived from deep learning and handcrafted radiomics features were evaluated for their predictive capabilities within a unified framework. The top 25 selected features are visualized in Figure 2, showing their relative contributions to the model's predictive performance.

Finally, the deep features from nnU-Net, the handcrafted radiomics features and the clinical data comprising the patient's age, resection status (GTR, STR, or NA), and the crucial clinical outcome of overall survival in days were concatenated to form a rich, multimodal feature vector for each patient. This comprehensive feature vector served as the input for subsequent SimCLR-based self-supervised representation learning and DDPG-based reinforcement learning modules designed for survival prediction.

3.4 | SimCLR Feature Representation

SimCLR (Simple Framework for Contrastive Learning of Visual Representations) is a self-supervised learning framework designed to learn meaningful and discriminative feature representations without requiring manual label annotation [31, 32]. This framework avoids the use of explicit class labels by employing data augmentations and a contrastive loss function, which effectively clusters similar samples closer in the feature space and distances dissimilar ones. The core principle is to produce two or more augmented views of the same input instance by applying a series of random transformations for images or perturbations for features. These augmented views are considered a positive pair, while other samples within the batch are regarded as negative examples. Each augmented view is processed through a base encoder, such as a Multi-Layer Perceptron (MLP) or a neural network, followed by a projection head that maps the encoded features into a latent representation space. The training of the model employs

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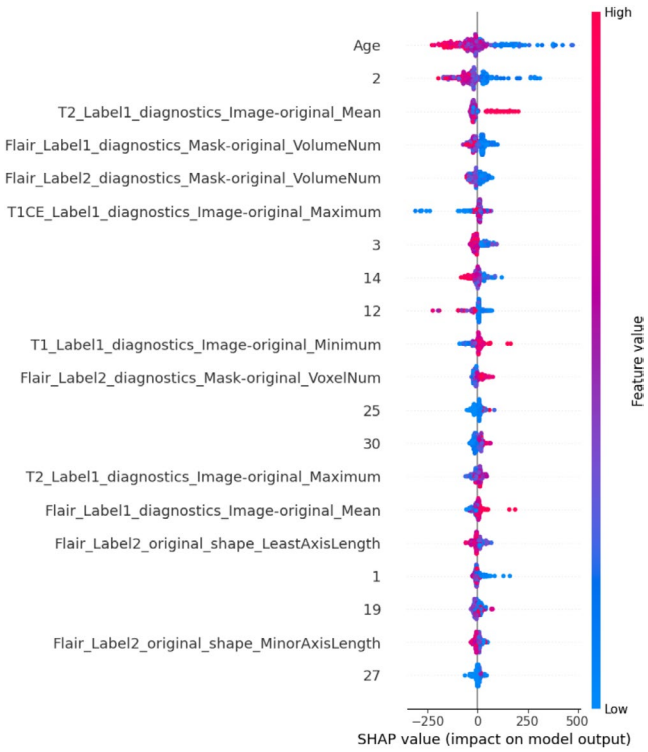

FIGURE 2 | Top 25 features selected by the BorutaShap algorithm from the fused deep and radiomics feature set. The SHAP value indicating each feature's contribution to survival prediction.

a contrastive loss function, specifically the Normalized Temperature-scaled Cross-Entropy Loss (NT-Xent), which encourages positive pairs to have aligned embeddings, while ensuring that negative pairs are pushed apart in the latent space. The architecture used in the SimCLR representation learning process is depicted in Figure 3.

SimCLR-based contrastive representation learning was independently performed for the nnU-Net deep features, handcrafted radiomics features, and the fused multimodal feature set comprising deep, radiomics, and clinical variables. This approach enabled the extraction of rich, discriminative representations from individual modalities as well as their combinations, facilitating robust downstream survival prediction.

To enhance the representation learning process, a series of randomized data augmentation operations including Gaussian noise addition, feature dropout, and scaling perturbations were applied to create multiple augmented versions of each patient's feature vector. These augmented feature pairs were then passed through a neural network equipped with a projection head composed of fully connected layers, mapping them into a lower-dimensional latent space suitable for contrastive learning. SimCLR optimized the NT-Xent loss with the formula as

$$L_{i,j} = -\log\left(\frac{\exp\left(\frac{\mathrm{sim}(z_i, z_j)}{\tau}\right)}{\sum_{k=1}^{2N} \mathbb{1}_{[k\neq i]} \exp\left(\frac{\mathrm{sim}(z_i, z_k)}{\tau}\right)}\right) \tag{1}$$

ALGORITHM 1 | Feature representation using SimCLR with MRI features.

Input: Fused feature dataset $X = \{x_1, x_2, \ldots, x_N\}$
for each feature vector $x_i \in X$ **do**
  Generate two augmented views:
    $\tilde{x}_i \leftarrow$ Apply Gaussian noise, feature dropout, and scaling perturbation to x_i
    $\tilde{x}_j \leftarrow$ Apply independent augmentations to x_i
  Pass augmented views through SimCLR encoder:
    $h_i = f_{enc}(\tilde{x}_i), h_j = f_{enc}(\tilde{x}_j)$
  Pass encoder outputs through projection head:
    $z_i = f_{proj}(h_i), z_j = f_{proj}(h_j)$
end for
Compute contrastive loss using NT-Xent:

$$\mathcal{L}_{i,j} = -\log \frac{\exp(\mathrm{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k\neq i]} \exp(\mathrm{sim}(z_i, z_k)/\tau)}$$

where:
  $\mathrm{sim}(z_i, z_j)$ is the cosine similarity between z_i and z_j
  τ is the temperature parameter
  $\mathbb{1}_{[k\neq i]}$ excludes i from the denominator
Update f_{enc} and f_{proj} parameters via backpropagation
Output: Trained encoder f_{enc} with learned representations

Here, $\mathrm{sim}(z_i, z_j)$ denotes the cosine similarity between the latent representations z_i and z_j of an augmented pair, τ is the temperature parameter, and N is the total number of samples in a mini-batch. This loss encourages augmented views of the same patient to cluster closely in the latent space while pushing representations of different patients farther apart, thereby learning modality-specific and fused discriminative features for survival prediction. The complete SimCLR-based representation learning procedure is summarized in Algorithm 1.

The PCA and t-SNE visualizations shown in Figure 4 compare raw fused features with SimCLR-transformed embeddings. In the PCA plots, raw features appear scattered with no clear structure, indicating a noisy feature space, whereas SimCLR features form tighter clusters, suggesting improved consistency and reduced redundancy. Similarly, t-SNE visualizations show that raw features produce diffuse distributions with high overlap, while SimCLR embeddings reveal clearer local clusters and better separability. Overall, SimCLR transforms the fused features into a more structured and compact representation, enhancing feature discriminability. This improved organization in the latent space provides a stronger foundation for downstream tasks, such as the Twin Critic DDPG survival prediction, promoting more stable learning and better generalization.

3.5 | Twin Critic Deep Deterministic Policy Gradient (Twin Critic DDPG) for Survival Prediction

The Twin Critic Deep Deterministic Policy Gradient is a reinforcement learning-based method applied for overall survival

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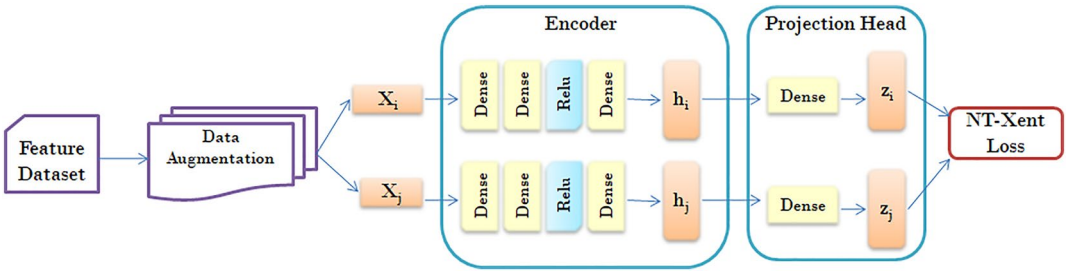

FIGURE 3 | SimCLR-based representation learning Architecture.

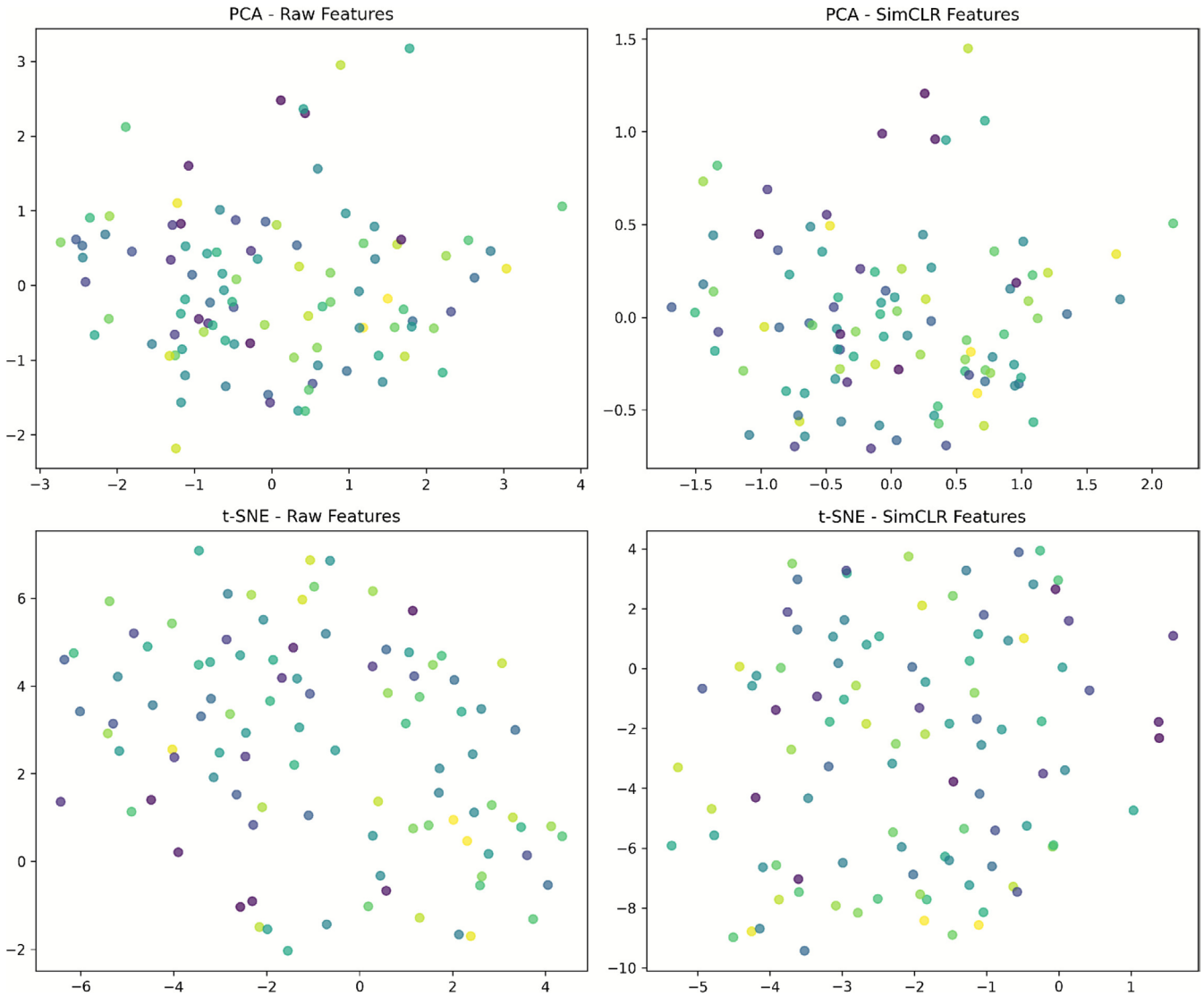

FIGURE 4 | PCA and t-SNE visualizations of raw fused features (left) and SimCLR embeddings (right). SimCLR produces more compact and separable feature representations, enhancing discriminability for downstream tasks.

prediction. It is a variant of the standard DDPG algorithm, enhanced by incorporating two critic networks to improve stability and reduce overestimation bias in Q-value estimates [33]. The architecture of the Twin Critic Deep Deterministic Policy Gradient (TD3) model used for continuous survival prediction is represented in Figure 5, showing the interaction between the actor, twin critics, and the environment.

The DDPG algorithm is a model-free, off-policy reinforcement learning technique designed for continuous action spaces. It is an extension of the Deterministic Policy Gradient (DPG) method, which adapts Q-learning for continuous environments [34, 35]. DDPG utilizes an actor-critic architecture, where the actor selects actions and the critic evaluates these actions by estimating Q-values for specific state-action pairs. The actor

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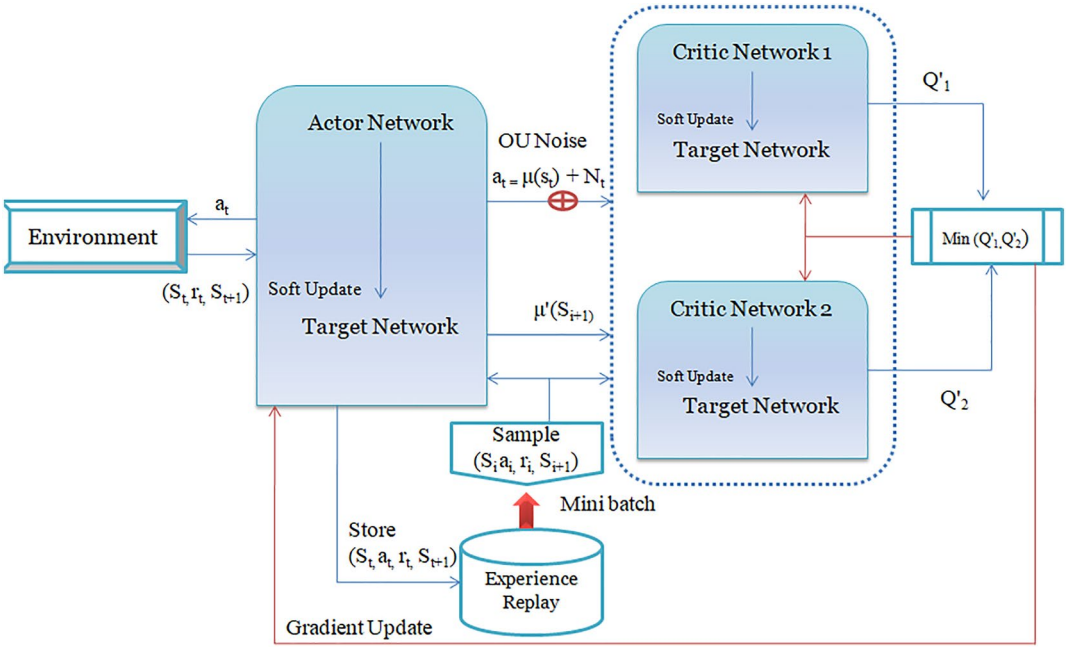

FIGURE 5 | Architecture of the Twin Critic DDPG agent with an actor network and the two independent critic networks.

network produces deterministic actions derived from the current state, while the critic network calculates Q-values based on the state-action pairs. The DDPG algorithm incorporates target actor and target critic networks, which are essentially copies of the original networks, and are updated at a slower rate to enhance training stability. This framework designed for predicting overall survival encompasses several key components.

3.6 | Environment

The environment represents the scenario for survival prediction involving a group of patients' data. It provides the agent with patient-specific feature vectors as the state, receives the agent's predicted survival time, computes a reward based on how close this prediction is to the true survival time, and then transitions to the next patient data. This setup simulates a continuous control environment specifically aimed at predicting survival outcomes for individual patients.

3.6.1 | State (*s*)

The state represents a fixed-length feature vector comprising patient-specific radiomics and deep features extracted from multi-modal MRI scans, along with selected clinical parameters. These features are obtained using SimCLR-based self-supervised learning, followed by fusion into a single representation. Each state corresponds to one patient and serves as the input for the reinforcement learning process.

3.6.2 | Action (*a*)

The action is a continuous-valued survival time prediction generated by the actor network. The temporally correlated Ornstein-Uhlenbeck (OU) noise is added to the action to encourage exploration during training.

3.6.3 | Reward (*r*)

The reward evaluates the precision of the agent's survival prediction by comparing the predicted survival time with the true survival time. This comparison is made through the negative absolute error between the two values, motivating the agent to decrease prediction errors. A lower absolute error corresponds to a higher (less negative) reward. This formulation provides a stable and interpretable learning signal that supports the efficient convergence of the reinforcement learning agent.

3.6.4 | Agent

The agent is responsible for learning the optimal survival prediction policy through interactions with the environment. The agent is trained using a Twin-Critic DDPG architecture consisting of the following components:

The patient-specific state is denoted $s_t \in \mathcal{S}$, obtained by fusing SimCLR-derived embeddings, radiomic descriptors, and nnU-Net segmentation features. The continuous action $a_t \in \mathbb{R}$ represents the predicted survival time at step t. After each action, the agent receives a scalar reward

$$r_t = -\,|\,a_t - y_t\,| \tag{2}$$

which penalizes the absolute deviation between the predicted survival time a_t and the ground-truth survival outcome y_t. Thus, higher rewards correspond to more accurate survival predictions.

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- **Actor Network:** The actor is a deterministic policy $\mu(s|\theta_\mu)$ parameterized by θ_μ that maps a patient state s_t to a continuous survival prediction a_t. Its optimization objective is to maximize the expected discounted return:

$$J(\theta_\mu) = \mathbb{E}_{s_0 \sim p_0}[V^\mu(s_0)], \qquad V^\mu(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s, a_t = \mu(s_t)\right] \tag{3}$$

where $\gamma \in [0, 1)$ is the discount factor. This encourages policies that produce survival predictions accurate across time. The deterministic policy gradient theorem [34] provides the update rule:

$$\nabla_{\theta_\mu} J(\theta_\mu) = \mathbb{E}_{s \sim \rho^\mu}\left[\nabla_a Q^\mu(s, a)|_{a=\mu(s)} \ \nabla_{\theta_\mu} \mu(s|\theta_\mu)\right] \tag{4}$$

where $Q^\mu(s, a)$ is the action-value function and ρ^μ is the state distribution induced by μ. In practice, this is approximated by sampling minibatches $\{s_i\}_{i=1}^N$ from the replay buffer:

$$\nabla_{\theta_\mu} J(\theta_\mu) \approx \frac{1}{N} \sum_{i=1}^{N} \nabla_a Q_{\theta_{Q_1}}(s_i, a)\Big|_{a=\mu(s_i)} \nabla_{\theta_\mu} \mu(s_i|\theta_\mu) \tag{5}$$

which is the minibatch-based deterministic policy gradient used to update the actor parameters.

- **Twin Critic Networks:** Two critic networks $Q_{\theta_{Q_1}}$ and $Q_{\theta_{Q_2}}$ are used to estimate the expected cumulative reward of state–action pairs:

$$Q_i(s_t, a_t) \equiv Q(s_t, a_t|\theta_{Q_i}), \qquad i \in \{1, 2\} \tag{6}$$

Each critic is trained by minimizing the mean-squared Bellman error. For critic k, the loss over a minibatch of size N is

$$\mathcal{L}_{Q_k}(\theta_{Q_k}) = \frac{1}{N} \sum_{i=1}^{N} \left(Q_{\theta_{Q_k}}(s_i, a_i) - y_i\right)^2, \qquad k \in \{1, 2\} \tag{7}$$

To avoid overestimation bias, the target value y_i is computed using the *clipped double-Q* method [36], where the minimum of the two target critics is taken:

$$y_i = r_i + \gamma \min\left(Q_{\theta_{Q_1^-}}(s_i', \mu(s_i'|\theta_{\mu^-})), \ Q_{\theta_{Q_2^-}}(s_i', \mu(s_i'|\theta_{\mu^-}))\right) \tag{8}$$

with (s_i, a_i, r_i, s_i') sampled from the replay buffer and $(\cdot)^-$ denoting target-network parameters. By using the more conservative estimate, Equation (6) mitigates the risk of the actor exploiting overly optimistic critic predictions.

- **Target Networks:** To stabilize training, target copies of the actor and critics are maintained. These target networks are updated by Polyak averaging:

$$\theta_{\mu^-} \leftarrow \tau\theta_\mu + (1-\tau)\theta_{\mu^-}, \qquad \theta_{Q_k^-} \leftarrow \tau\theta_{Q_k} + (1-\tau)\theta_{Q_k^-}, \quad k \in \{1, 2\} \tag{9}$$

where $\tau \ll 1$ (e.g., 10^{-3}). This slow update prevents oscillations and improves stability during bootstrapped value estimation.

Together, Equations (2–9) define the learning dynamics of the SimCLR-informed Twin-Critic DDPG agent: the actor generates survival predictions, the twin critics evaluate them, the target networks stabilize training, and the reward penalizes prediction error. This framework leverages strong feature representations with stable value estimation to achieve accurate and robust survival prediction. This SimCLR-based feature representation with a Twin Critic DDPG algorithm for survival prediction is explained in Algorithm 2.

The performance of the proposed framework is assessed using several evaluation metrics, including Concordance Index (C-Index), Mean Squared Error (MSE), and Mean Absolute Error (MAE). These metrics collectively quantify both the predictive accuracy and the reliability of the model in survival prediction tasks.

3.6.5 | Evaluation Metrics

The assessment for the segmentation performance is evaluated using the DSC and IoU. The DSC is defined as:

$$\text{DSC} = \frac{2\,|A \cap B|}{|A| + |B|} \tag{10}$$

where A is the set of voxels in the predicted segmentation, and B is the set of voxels in the ground truth mask. Dice values range from 0 to 1, with 1 indicating perfect overlap.

The IoU quantifies the ratio of the overlapping area to the total area of both the predicted and true regions. Similar to Dice, IoU values range from 0 to 1, where higher values denote better segmentation performance:

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|} \tag{11}$$

The C-Index is a widely used metric in survival analysis to compute the concordance between the predicted and actual survival times. A higher C-Index value (closer to 1) indicates better predictive discrimination ability:

$$\text{C-Index} = \frac{\text{Number of concordant pairs}}{\text{Number of comparable pairs}} \tag{12}$$

The MSE and MAE evaluate the average prediction error in terms of squared and absolute differences, respectively:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2 \tag{13}$$

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ALGORITHM 2 | Twin-Critic DDPG with SimCLR feature representations.

Initialize:
Initialize actor network $\mu_{\theta^{\mu}}$, twin critic networks $Q_{\theta^{Q1}}, Q_{\theta^{Q2}}$
Initialize target networks $\mu_{\theta^{\mu'}}, Q_{\theta^{Q1'}}, Q_{\theta^{Q2'}}$ with weights equal to their respective online networks
Initialize replay buffer D
Set hyperparameters: learning rates α, β, discount factor γ, soft update rate τ, OU noise parameters $\mathcal{N}_t$
for episode = 1 to M **do**
Reset environment, set initial state s_0 using pre-encoded SimCLR representations
for step = 1 to T **do**
Select action (predicted survival time) with exploration noise:

$$a_t = \mu\left(s_t \mid \theta^{\mu}\right) + \mathcal{N}_t$$

Compute reward:

$$r_t = - \mid a_t - y_t \mid$$

Apply action a_t, observe reward r_t and next state s_{t+1}
Store transition $\left(s_t, a_t, r_t, s_{t+1}\right)$ in replay buffer D
Sample minibatch of N transitions from D
Compute target Q-value using twin target critics:

$$y_i = r_i + \gamma \min\left(Q_{\theta^{Q1'}}\left(s_{i+1}, a_i'\right), Q_{\theta^{Q2'}}\left(s_{i+1}, a_i'\right)\right)$$

Update both critic networks by minimizing:

$$L^{Qk}\left(\theta^{Qk}\right) = \frac{1}{N}\sum_{i=1}^{N}\left(Q_{\theta^{Qk}}\left(s_i, a_i\right) - y_i\right)^2, \quad k = 1, 2$$

Update actor network using the policy gradient:

$$\nabla_{\theta^{\mu}} J \approx \frac{1}{N}\sum_{i=1}^{N} \nabla_a Q_{\theta^{Q1}}\left(s_i, \mu\left(s_i \mid \theta^{\mu}\right)\right) \nabla_{\theta^{\mu}} \mu\left(s_i\right)$$

Update target networks:

$$\theta^{Qk'} \leftarrow \tau\theta^{Qk} + (1-\tau)\theta^{Qk'}, \quad k = 1, 2$$

$$\theta^{\mu'} \leftarrow \tau\theta^{\mu} + (1-\tau)\theta^{\mu'}$$

end for
end for

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N} \mid y_i - \hat{y} \mid \tag{14}$$

4 | Results and Discussion

4.1 | Evaluation of the Image Segmentation

In this study, multimodal MRI sequences including FLAIR, T1, T1CE, and T2 images from the BraTS2020 dataset were provided as input to four well-known segmentation models such as UNet, VNet, SwinUNETR, and nnU-Net. These models were trained exclusively on the training set, ensuring that no test set images were used during model development. The formulae used to compute DSC and IoU are given in (9) and (11), respectively. The detailed segmentation results are summarized in Table 1. Among the models, nnU-Net consistently outperformed others, achieving the highest DSC and IoU scores across all tumor subregions reflecting its adaptability and auto-configuration capability for biomedical imaging tasks. Since nn-U-Net got the superior results, the deep features are extracted from nnU-Net segmentations and fused with handcrafted radiomics features. All radiomics and deep features were extracted after the training/test split, ensuring complete isolation of the test set and preventing information leakage. BorutaShap selected features added with clinical features were passed through a SimCLR-based self-supervised module to enhance representation quality for survival prediction.

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TABLE 1 | Segmentation performance of four models.

Model	Dice TC	Dice WT	Dice ET	Average dice	Average IoU
UNet	0.8562	0.8461	0.8528	0.8517	0.7405
VNet	0.8881	0.8962	0.8592	0.8812	0.7848
SwinUnetR	0.8940	0.8928	0.8813	0.8894	0.7910
nnU-Net	0.8981	0.8964	0.8827	0.8924	0.7974

TABLE 2 | Hyperparameter tuning ablation study of Twin-Critic DDPG.

γ	τ	Actor LR	Critic LR	MSE (fused features)
0.90	0.01	0.001	0.001	40 850.42
0.95	0.01	0.001	0.001	35 473.78
0.95	0.05	0.001	0.002	**31 062.36**
0.99	0.05	0.001	0.002	34 002.11
0.95	0.10	0.0005	0.002	36 512.67
0.95	0.05	0.0005	0.001	37 890.30

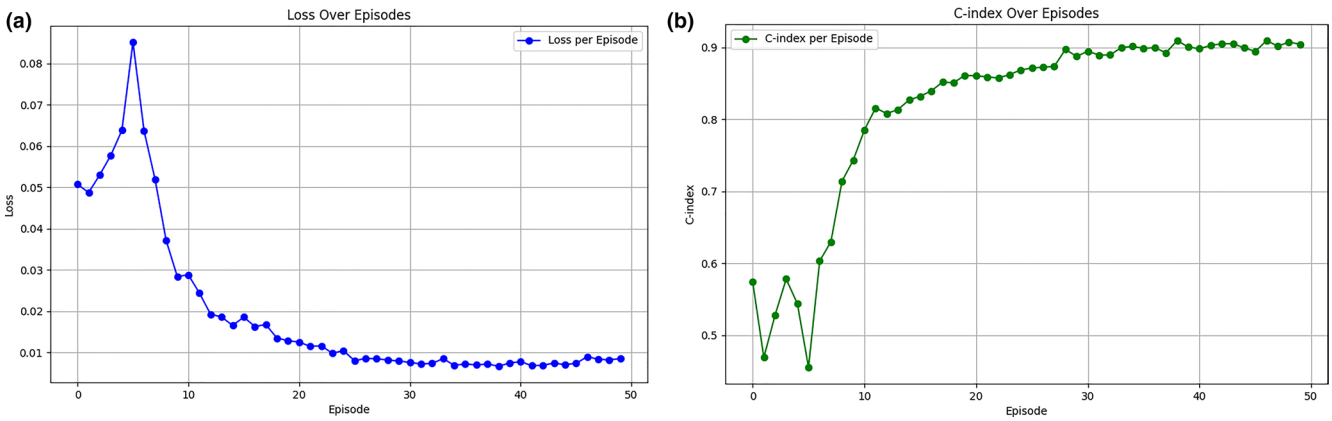

FIGURE 6 | Training performance of the SimCLR-Twin Critic DDPG model during training.

4.2 | Evaluation of the Survival Prediction

The SimCLR-represented features were used as input to the proposed Twin Critic DDPG framework. The experiments were conducted on a system equipped with a 12th Gen Intel(R) Core(TM) i7-12700 processor and a 64-bit operating system. To determine the optimal hyperparameter configuration for the proposed Twin-Critic DDPG framework, an ablation study was conducted by varying key parameters such as the discount factor (γ), soft update coefficient (τ), and learning rates for the actor and critic networks. As shown in Table 2, the best performance in terms of mean squared error (MSE) on fused features was achieved with $\gamma = 0.95$, $\tau = 0.05$, actor learning rate = 0.001, and critic learning rate = 0.002. These results indicate that moderate discounting ($\gamma \approx 0.95$) and moderate soft updates improve stability and generalization on this small dataset, while values of γ very close to 1 increased variance and degraded performance. Furthermore, the combination of lower discount factors with the negative absolute error reward function consistently promoted stable convergence, which is particularly important when working with limited medical datasets.

The model was trained separately using radiomics features, nnU-Net deep features, and their fused features. Survival prediction performance was evaluated using C-Index, MSE and MAE formulae as given in (12), (13), and (14), respectively. Figure 6 shows the training curves illustrating the convergence of MSE and C-index of the proposed framework. As illustrated in Table 3, the baseline DDPG and DeepSurv models achieve reasonable performance but exhibit higher error and lower concordance on the small, high-dimensional dataset. Incorporating self-supervised representations with SimCLR improves DeepSurv, yielding better generalization and higher concordance. The reinforcement learning variant Twin Critic DDPG further reduces error and improves stability, while our proposed SimCLR-Twin Critic DDPG consistently achieves the

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TABLE 3 | Evaluation results of SimCLR-Twin Critic DDPG with baseline methods.

	Radiomics features			nnU-Net			Fused features		
Methods	**MSE**	**MAE**	**C-Index**	**MSE**	**MAE**	**C-Index**	**MSE**	**MAE**	**C-Index**
DeepSurv	45 210.32	152.11	0.72	44 890.54	136.67	0.74	44 334.21	118.45	0.74
SimCLR + DeepSurv	43 765.18	142.29	0.76	42 502.46	129.84	0.76	42 103.77	122.36	0.76
DDPG	42 970.85	149.85	0.80	41 501.21	126.23	0.77	40 068.89	107.61	0.79
Twin Critic DDPG	34 913.44	119.24	0.82	33 995.94	98.49	0.81	33 048.33	99.20	0.85
SimCLR-Twin Critic DDPG	32 073.14	105.20	0.83	32 878.50	94.30	0.85	**31 062.36**	**96.73**	**0.87**

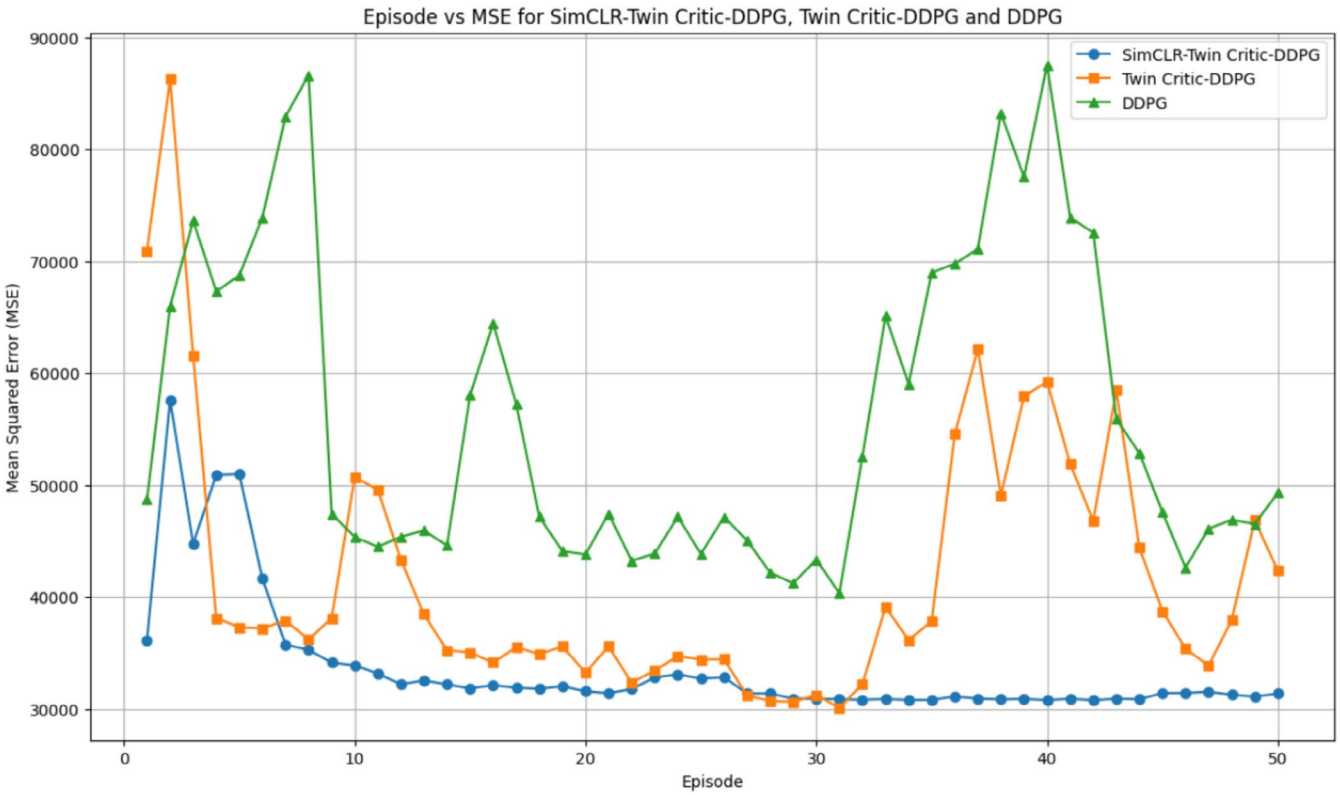

FIGURE 7 | Comparison of mean squared error (MSE) across different models: Standard DDPG, Twin Critic DDPG, and the proposed SimCLR-Twin Critic DDPG. The proposed framework consistently achieves lower prediction error, indicating improved survival prediction performance using fused feature representations and contrastive pretraining.

best results with the lowest MSE of 31 062.36, an MAE of 96.73, and the highest C-Index of 0.87. These findings demonstrate that combining SimCLR-based representation learning with a Twin Critic reinforcement learning agent yields robust and accurate survival predictions, effectively outperforming both traditional and deep survival baselines. Figure 7 presents a comparison of MSE values across SimCLR-Twin Critic DDPG, standard DDPG, and Twin Critic DDPG models, demonstrating that our proposed framework achieves consistently lower prediction error throughout training.

The BraTS 2019 dataset was used for external validation. The results of working on BraTS 2019 via SimCLR–Twin Critic DDPG model were similar to the results of working on BraTS 2020, as shown in Table 4. According to the results, the fused feature set achieved the best performance once again. This confirms that the framework is robust and generalizable to independent cohorts.

TABLE 4 | External validation results of SimCLR–Twin Critic DDPG on BraTS 2019 dataset.

Features	MSE	MAE	C-Index
Radiomics features	32 450.67	107.32	0.82
nnU-Net features	32 925.28	95.29	0.84
Fused features	**31 156.56**	**96.10**	**0.86**

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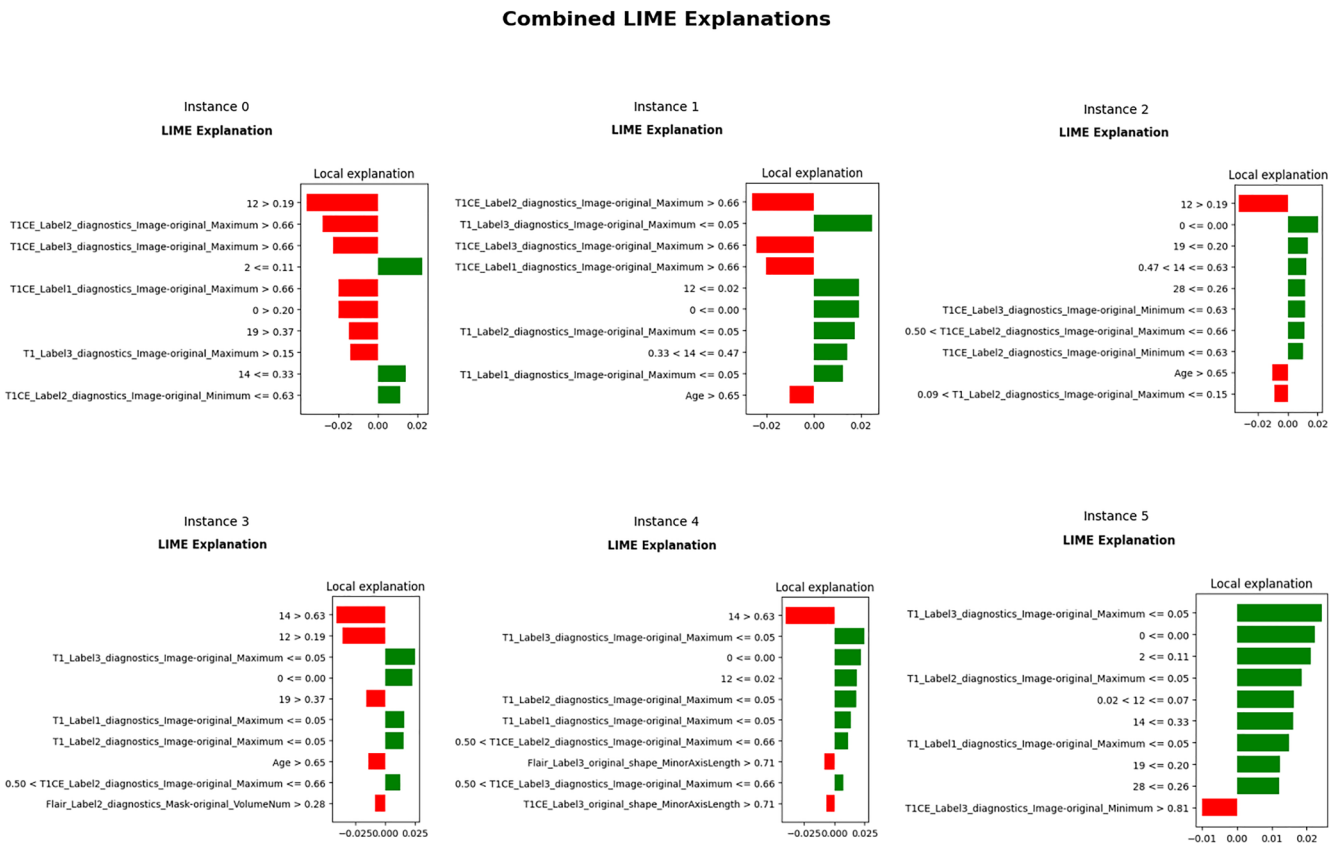

FIGURE 8 | Local interpretability analysis using LIME for six patient instances. Each subplot shows the ten most influential features for the individual survival prediction. Green bars indicate features with positive influence, while red bars indicate features with negative influence on the survival prediction.

4.3 | LIME Interpretation for Survival Prediction

LIME is a model-agnostic explainer technique, assessing the influence of specific features on predictions and offering insights into the decision-making process of the model [37]. Figure 8 visually represents six distinct patient instances providing critical insight into the decision-making process of the survival prediction model. Each subplot corresponds to a single patient and highlights the 10 most influential features for that individual prediction. In this analysis, radiomics, nnU-Net, and clinical features all play significant roles in shaping the survival prediction.

In the visualization, green bars denote features that positively impact survival duration, whereas red bars indicate features that negatively affect it. The length of each bar reflects the magnitude of its influence. Across these patients, both radiomic and nnU-Net-derived features frequently dominate the predictions, particularly those related to maximum intensity values in post-contrast T1 sequences (e.g., `T1CE_Label2_diagnostics_Image-original_Maximum`) and baseline T1 sequences. Features reflecting tumor size and intensity (e.g., necrotic core and enhancing tumor regions captured by nnU-Net features) as well as patient age contribute meaningfully to the predictions. This consistency supports the clinical interpretability and reliability of the model, allowing clinicians to understand how imaging-derived and patient-specific factors influence individualized survival estimates.

TABLE 5 | Performance comparison of proposed method with the existing methods.

Authors	Method	MSE
Ali et al. [38]	RF + Grid Search	392 963.18
Baid et al. [20]	Multilayer Perceptron	338 219.37
Zhu et al. [19]	Support Vector Regression	139 571.96
Tran et al. [17]	Light GBM with a majority voting	97 216.34
Huang et al. [21]	Random Forest	96 470.98
Rajput et al. [18]	Random Forest Regressor	79 826.24
Akbar et al. [22]	MobileNet + Linear SPM	78 374.17
Chen et al. [16]	SVC with Ensemble Regressor	43 674.49
Proposed	SimCLR-Twin Critic DDPG	31 062.36

4.3.1 | Comparison With Existing Methods

To evaluate the efficacy of the proposed SimCLR-Twin Critic DDPG framework, its Mean Squared Error (MSE) was benchmarked against several recent studies for the task of overall survival prediction, as illustrated in Table 5. Our framework achieved a significantly reduced MSE of 31 062.36, indicating

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superior predictive accuracy. This performance surpasses that of traditional machine learning algorithms such as LightGBM, Support Vector Classification (SVC), Random Forest Regressor (RFR), Ridge Regression, and Support Vector Regression (SVR). Furthermore, the proposed model also outperforms advanced deep learning architectures, including Multi-Layer Perceptron (MLP) and MobileNet-SPM. The comparatively higher MSE values reported by these methods emphasize the advantage of integrating self-supervised feature learning via SimCLR with a Twin Critic DDPG reinforcement learning agent. Such results validate the effectiveness and generalization capability of our framework across diverse patient data distributions.

5 | Conclusion

Accurate survival prediction in glioblastoma is vital for personalized treatment planning but remains challenging due to limited annotated data, tumor heterogeneity, and dynamic clinical conditions. There is a clear need for models that are adaptive, data-efficient and interpretable. This study presents a novel framework that synergistically combines SimCLR-based self-supervised learning with Twin-Critic Deep Reinforcement Learning for continuous survival prediction. The proposed approach demonstrates superior performance on benchmark datasets, establishing a strong foundation for adaptive and generalizable AI-driven prognosis. Experimental results confirm its robustness and effectiveness, highlighting its clinical potential for improving survival prediction in glioblastoma patients. A key limitation is that this work was validated only on the BraTS dataset, and future work will include external validation on independent cohorts and further enhancement of the framework through ensemble strategies.

Author Contributions

Renugadevi M. led the conceptualization, implementation, and manuscript drafting. Ramkumar K. and Narasimhan K. supervised the study and provided overall guidance. Raju N., Adalarasu K. and Surya Prasath offered theoretical insights and critical revisions. All authors reviewed and approved the final manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The BraTS2020 dataset used for this study is available at https://www.med.upenn.edu/cbica/brats2020/data.html. The BraTS2019 dataset is available at https://www.med.upenn.edu/cbica/brats2019/data.html.

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